Notes on Nuclear Astrophysics

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මෙම සටහන් පෙළ සකස් කර ඇත්තේ න්‍යෂ්ටික විද්‍යා අංශයේ න්‍යෂ්ටික තාරකා විද්‍යා මොඩියුලය සඳහා සහය දැක්වීමටයි. මෙම මොඩියුලය මුලින් ලබන වසර සඳහා සැලසුම් කර තිබුණද, ඔබගේ ඉල්ලීම් ඉටු කිරීම සඳහා විෂය නිර්දේශය වේගවත් කර ඇත. එහි ප්‍රතිඵලයක් ලෙස, ඔබගේ සාර්ථකත්වයට අවශ්‍ය සම්පත් ඔබට ලැබෙන බව සහතික කිරීම සඳහා මම මෙම ද්‍රව්‍ය කෙටි කාලයක් තුළ සකස් කළෙමි.

ශ්‍රී ලංකාවේ පර්යේෂණ යටිතල පහසුකම් සහ මානව සම්පත් සීමිත වීම නිසා න්‍යෂ්ටික තාරකා විද්‍යාව බොහෝ විට සීමා වී ඇත. කෙසේ වෙතත්, ගෝලීය විද්‍යාත්මක ප්‍රජාව විසින් පෙන්නුම් කර ඇති පරිදි, අප දියුණු වන්නේ සහයෝගී විද්‍යාව තුළින්ම ය. මගේ ඉලක්කය වන්නේ මෙම විද්‍යාව සැමට ප්‍රවේශ විය හැකි ලෙස සැකසීමයි. විද්‍යාව තරඟයක් නොව සහයෝගී ප්‍රයත්නයක් බව පිළිගන්නා ලෙස මම ඔබට ආරාධනා කරමි. ඔබ පුළුල් න්‍යෂ්ටික තාරකා විද්‍යා ප්‍රජාව නියෝජනය කිරීමට සහ ශ්‍රී ලංකාවේ අධ්‍යයන ක්ෂේත්‍රයට නැවත දායක වීමට වර්ධනය වනු ඇතැයි මම බලාපොරොත්තු වෙමි. තරු තුළ සිදුවන සංකීර්ණ න්‍යෂ්ටික සංස්ලේෂණ ක්‍රියාවලීන් ජීවයට අවශ්‍ය මූලද්‍රව්‍ය නිර්මාණය කරනවා සේම, ඉගෙනීමට ඔබ දක්වන කැපවීම අපේ ජාතියේ විද්‍යාවේ අනාගතය ගොඩනඟනු ඇත.

මෙම සටහන් මෙම මොඩියුලයේ ඔබගේ සාර්ථකත්වයට අවශ්‍ය අතිරේක සටහන් සපයයි. ඔබ සැම මා සමඟ සම්බන්ධ වීම ගැන මම සතුටු වෙමි. මෙම මාලාව ස්ට්‍රාස්බර්ග් තාරකා විද්‍යා නිරීක්ෂණාගාරයේ මාස්ටර් 2 වැඩසටහනෙන් ආභාෂය ලබා ඇත. ඔවුන්ගේ වටිනා මගපෙන්වීම සඳහා එහි සිටින ගුරුවරුන්ට සහ පර්යේෂකයන්ට මගේ විශේෂ ස්තුතිය පුද කරමි.

මෙම සටහන් ගැඹුරු න්‍යායාත්මක සංකල්ප හැකි තරම් සරලව පැහැදිලි කිරීමට උත්සාහ කරයි. මෙය සාම්ප්‍රදායික දේශන සඳහා ආදේශකයක් නොව, අතිරේක සම්පතකි. මෙය මෙම ව්‍යාපෘතියේ පළමු අනුවාදය බැවින්, එය තවදුරටත් සංවර්ධනය කිරීමට උදවු වන ලෙස ඔබගේ ප්‍රතිපෝෂණ සහ සහයෝගය මා ඉතා නිහතමානීව ඉල්ලා සිටිමි.

ඔබගේ අධ්‍යයන කටයුතු සාර්ථක වේවායි ප්‍රාර්ථනා කරමි

Chapter 1

Aspects of Astronomy & Astrophysics

e is one of the most photogenic examples of the many turbulent stellar nurseries the NASA/ESA Hubble Space Telescope has observed during its 30-year lifetime. The portrait features the giant nebula NGC 2014 and its neighbour NGC 2020 which together form part of a vast star-forming region in the Large Magellanic Cloud, a satellite galaxy of the Milky Way, approximately 163 000 light-years away.

0.1 Section 1

Astronomy is a natural science dedicated to the study of celestial objects and phenomena beyond Earth's atmosphere. Unlike chemists or biologists, astronomers cannot bring a star or a galaxy into a laboratory to conduct controlled experiments. Instead, the universe itself is our laboratory. Because we cannot manipulate celestial bodies, our work relies heavily on observation rather than traditional experimentation. Most of our understanding comes from analyzing the light—or photons—that travels across space to reach our telescopes. While visible light was our only window into the cosmos for centuries, modern technology has allowed us to expand our reach. We now use specialized instruments to detect Radio waves, X-rays, and Gamma rays We have developed the ability to detect cosmic rays, neutrinos, and gravitational waves, providing us with entirely new ways to "listen" to the universe.

A fundamental principle of astronomy is that the speed of light is finite. Light takes time to travel through the vast distances of space. Consequently, when we look at distant stars and galaxies, we are not seeing them as they are today; we are seeing them as they were when that light first began its journey toward us. In effect, every telescope is a time machine, allowing us to observe the history of the universe.

Until 1609 (Galileo's first telescope), observations were done with the naked eye. Until the 1940s, only wavelengths near the optical (visible) were used, then radio astronomy started. And later, other domains in the spatial era.

Astrophysics evolved from this classical astronomical foundation during the 1940s and 1950s, shifting the focus toward the physical study of the universe and its constituent parts. This transformation was propelled by two groundbreaking discoveries one of course belogn to the modern nuclear astrophysics domain

- **Stellar Energy:** The discovery that nuclear fusion of hydrogen is the power source for stars. Remarkably, scientists like Arthur Eddington and Cecilia Payne-Gaposchkin theorized the core mechanisms of stars well before the specific physics of nuclear fusion were fully understood.
- **An Expanding Universe:** The observation that the universe is not static but is actively expanding.

Today, astrophysics integrates these theoretical foundations with advanced detector technology to provide a comprehensive understanding of astronomical objects, allowing us to explain not just where celestial bodies are, but how they live, evolve, and die.

1

Chapter 2

Nomenclature

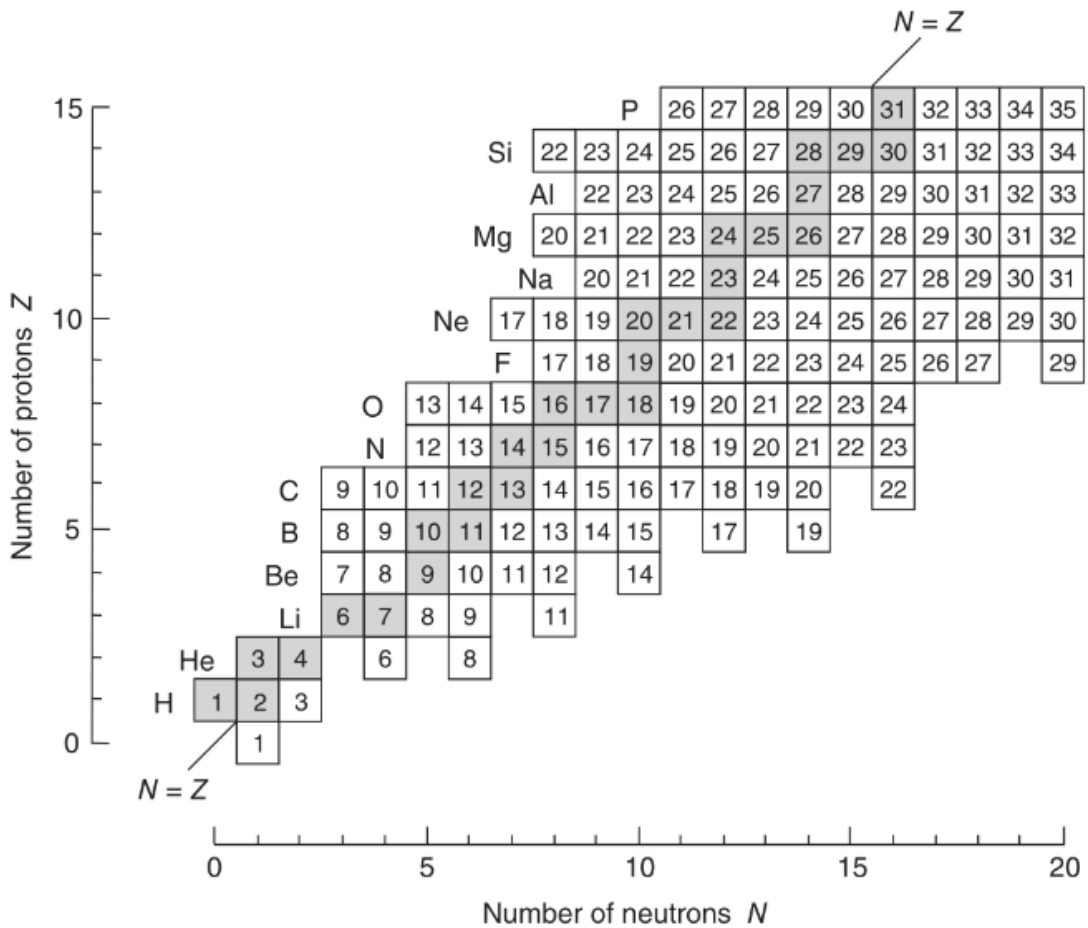
An atomic nucleus is a compact, dense central region of an atom composed of protons and neutrons, which are collectively referred to as nucleons. The composition of a nucleus is defined by two primary parameters: the atomic number (Z), representing the number of protons, and the neutron number (N), representing the number of neutrons. The sum of these two, A=Z+N, is known as the mass number (or nucleon number), and any individual nuclear species defined by this specific composition is called a nuclide. Nuclides are categorized based on their proton and neutron relationships, as detailed in the accompanying table, which distinguishes between isotopes, isobars, and isotones.

Category	Definition
Isotopes	Nuclides with the same number of protons (Z) but a different number of neutrons (N).
Isobars	Nuclides with the same mass number (A) but different numbers of protons and neutrons.
Isotones	Nuclides with the same number of neutrons (N) but different numbers of protons (Z).

For this course, a nuclide is represented by the symbol ${}_Z\mathrm{X}_N$, where X is the element symbol.

1.1 The Chart of the Nuclides

Nuclides are systematically organized in a two-dimensional plot known as the chart of the nuclides. In this representation, the number of neutrons (N) and the number of protons (Z) are plotted on the horizontal and vertical axes, respectively. Each square on the grid corresponds to a specific nuclide, characterized by its unique nuclear physical properties. For an interactive exploration of this landscape, one may refer to the online chart of the nuclides.



1.1: Section of the chart of the nuclides, showing the lightest species with $Z \leq 15$ and $N \leq 20$. The shaded squares represent stable nuclides, while the open squares correspond to unstable nuclides with half-lives in excess of 1 ms. The only exceptions are the nuclides ^{8}Be and ^{9}B, which have considerably shorter half-lives. No stable nuclides exist with a mass number of $A = 5$ or 8.

1.2 Stellar Physics

1.2.1 The Sun: Our Benchmark

Steller physics is the subdomain of astronomy which study the star. The stellar physics is important in astrophysics becuase stars are long-lived astronomical objects that provide most of the light in the universe. — Studying stars for their own sake — Studying stars as members of stellar systems — Stars synthesise heavy elements — Stars generate energy for the ISM — Stars as objects to help unravel the structure of the Milky Way — Stars as objects to calibrate cosmological distances

The Sun serves as the perfect archetype for this study. As a nuclear astrophysics student sun is important becuase it is the closest natural fusion the Sun currently generates energy by fusing hydrogen into helium in its core. It has been in this state of hydrostatic equilibrium for approximately 4.5 billion years and will continue to be for billions more. By understanding the Sun—its mass, luminosity, surface temperature, and internal nuclear processes—we gain the essential key to unlocking the physics of all other stars. Figure (ref sun) show the sun seen

Table (sun) illustrates key characterstics for siun.. As a nucleares sicne student you should at least know the age, mass and radium os the sun and surfCE density